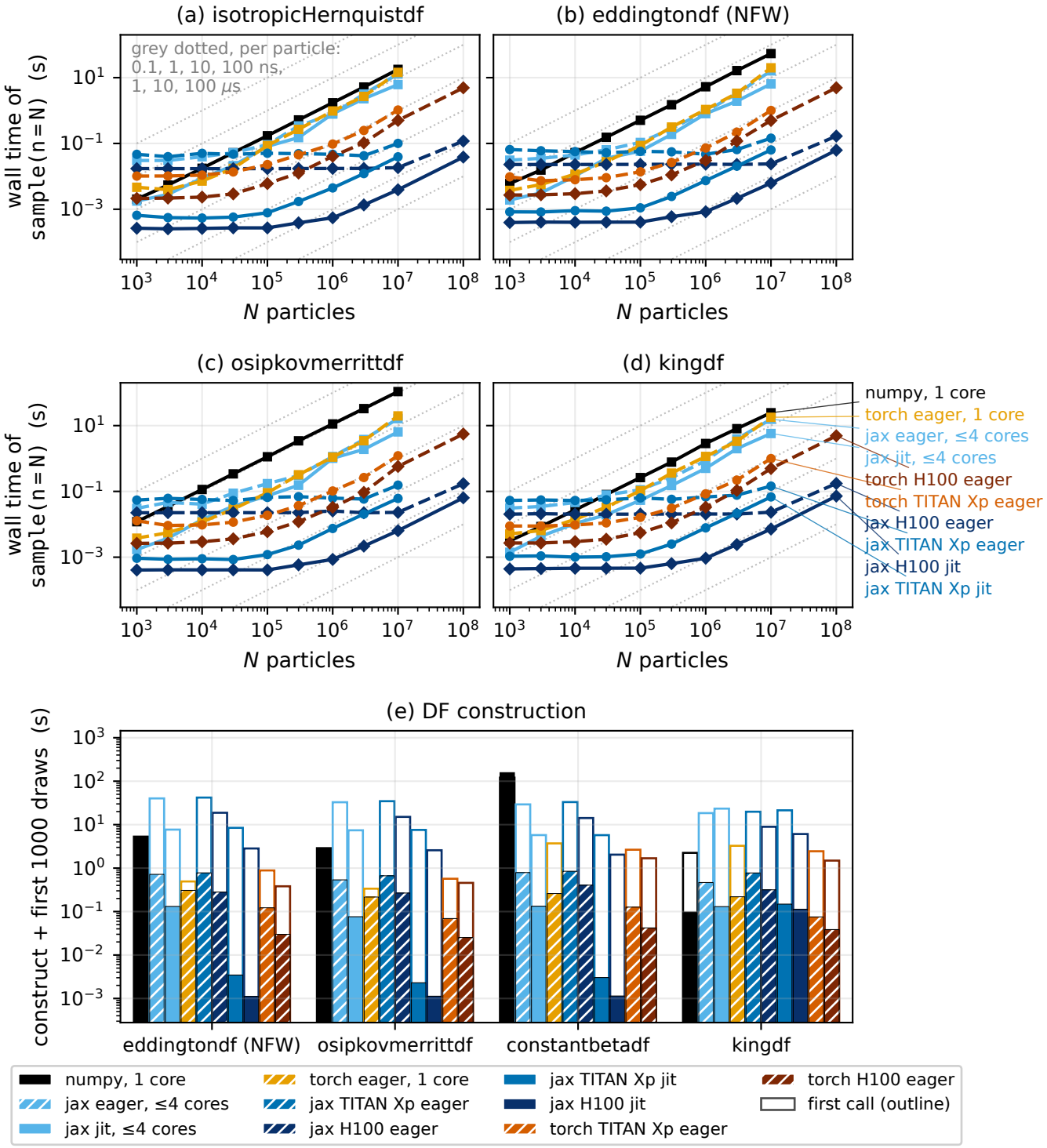
(a) isotropicHernquistdf
(b) eddingtondf (NFW)
(c) osipkovmerrittdf
(d) kingdf
(e) DF construction
grey dotted, per particle:
0.1, 1, 10, 100 ns,
1, 10, 100 μs
wall time of sample(n=N) (s)
N particles
numpy, 1 core
torch eager, 1 core
jax eager, ≤4 cores
jax jit, ≤4 cores
torch H100 eager
torch TITAN Xp eager
jax H100 eager
jax TITAN Xp eager
jax H100 jit
jax TITAN Xp jit
construct + first 1000 draws (s)
eddingtondf (NFW)
osipkovmerrittdf
constantbetadf
kingdf
numpy, 1 core
jax eager, ≤4 cores
jax jit, ≤4 cores
torch eager, 1 core
jax TITAN Xp eager
jax H100 eager
jax TITAN Xp jit
jax H100 jit
torch TITAN Xp eager
torch H100 eager
first call (outline)